

Craters as Constellations: Adapting Star Identification to Lunar Crater Identification

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Abstract—Crater identification (CID) is structurally analogous to star identification (SID) since both must match an unordered, catalog-incomplete observation set using only non-dimensional quantities when no state information is available. However, only a limited number of studies have adapted pre-2017 SID algorithms to the CID problem. Therefore, we selected two recent approaches, the deterministic TETRA and the learning-based SIGNN, and developed corresponding CID methods, named CITRA and CIGNN. Unlike most existing approaches, which primarily rely on the geometric configuration among craters, our methods also exploit crater-specific geometric information. CITRA extends the pattern key using crater rim-area ratios, whereas CIGNN uses the relative area and eccentricity of imaged craters as node features. Both proposed methods are evaluated on the QuickMap dataset at three off-nadir angles, together with Perspective Cone Alignment (PECAN) as a baseline CID algorithm. Although the proposed methods achieve lower identification accuracy than PECAN, CIGNN requires only approximately 2.1 ms per frame on an RTX 5090, making it significantly faster than PECAN running on a single CPU core. These results suggest that modern SID techniques can be effectively adapted to CID and have strong potential for real-time lunar lander navigation.

I. INTRODUCTION

Since GNSS and continuous ground support are unavailable in lunar missions, orbiters and landers must recover their absolute pose onboard. Most pose recovery methods require landmarks, and impact craters are the most abundant and stable landmarks for this purpose. Therefore, recognition of observed crater patterns against a global database without any state information, the lost-in-space (LIS) CID, is the basis of crater-based terrain relative navigation (TRN) [1], [2].

SID and CID share the same structure. In a LIS scenario, the altitude and hence the image scale are unknown, so only non-dimensional quantities are usable [3] exactly as in a star tracker. Also, both must match an unordered, catalog-incomplete observation set. This has motivated transfers of SID concepts to CID. However, a closer examination reveals that CID is considerably more challenging, mainly because geometric properties that are invariant in SID can vary with the camera pose in CID. Accordingly, previous studies have attempted to adapt SID methods so that they can be applied to CID, but several limitations remain. Xu et al. [4] applied

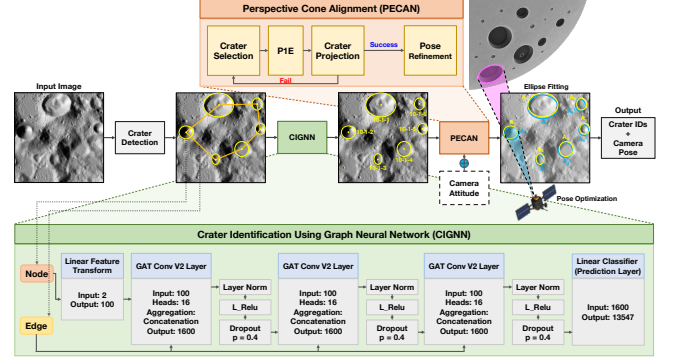


Fig. 1: CIGNN pipeline. It consists of three components: crater detection, CID, and pose optimization.

the Pyramid algorithm to CID by using projective invariants; however, the resulting method remained highly sensitive to noise introduced during the crater detection stage. Hanak et al. [3] introduced scale-invariant features based on crater triangles, but triangle angles and relative edge geometry vary with off-nadir viewing, making the method difficult to apply under large off-nadir angles.

To address the limitations of existing works, we exploit crater-specific features to adapt two recent approaches of SID: the deterministic TETRA [5] and the learning-based SIGNN [6]. *CIGNN* (*Crater Identification via Graph Neural Network*) uses scale-invariant quantities as node and edge features and is trained on datasets containing various off-nadir angles to learn robustness to projective changes. In addition, the training dataset is augmented to improve robustness to crater detection errors. On the other hand, *CITRA* (*Crater Identification via teTRA*) rectifies off-nadir images with a known camera attitude, and uses the ratios of edge length and crater area as pattern keys of a hash-table-based database. Our methods are used to investigate the fundamental compatibility and differences between the two landmark-identification problems.

II. METHOD

A. Crater Identification via Graph Neural Network (CIGNN)

Fig. 1 illustrates the workflow of the proposed CIGNN framework. Given an image of the lunar surface, crater rims are first extracted through crater detection. Since CIGNN requires not only the crater centers but also the ellipse parameters of the crater rims, a crater detection algorithm capable of fitting ellipses is required. However, the crater detection stage is bypassed in order to evaluate and analyze the CID performance independently. Edge connection is the following step before the Graph Neural Network, which

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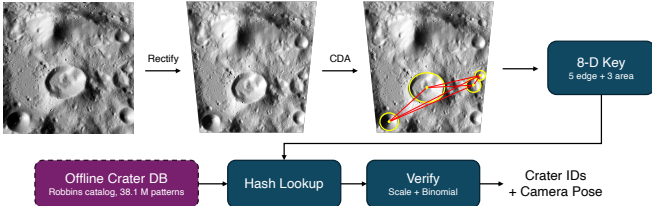


Fig. 2: CITRA pipeline. Rectification restores the similarity relation that the edge- and size-ratio key requires. Four-crater patterns are keyed by five edge ratios and three rim-area ratios, looked up in an offline hash table, and verified by reprojecting the catalog.

predicts the crater IDs and their associated confidence scores. Subsequently, the detected craters are visited in descending order of confidence, and Perspective Cone Alignment (PECAN) [7] is executed for each candidate. If the crater ID of an imaged crater is already inferred, there is no need to search the entire catalog, reducing the computational complexity from $\mathcal{O}(N)$ to $\mathcal{O}(1)$.

The original PECAN is accurate but costly. Assuming known attitude, it recovers the camera position through the Perspective-One-Ellipse (PIE) formulation of [8], projects the catalog into the image, establishes correspondences by nearest-neighbor matching on ellipse parameters, and refines the full 6-DoF pose.

Within CIGNN, we employ a three-layer Graph Attention Network (GAT) to process the crater graph. The network can aggregate information not only from each crater’s immediate neighbors but also from nodes within its 3-hop neighborhood. As node features, we use the relative crater area and eccentricity, defined as $A_{rel,i} = \frac{A_i - \min_{j \in \mathcal{N}(i)}(A_j)}{\max_{j \in \mathcal{N}(i)}(A_j) - \min_{j \in \mathcal{N}(i)}(A_j)}$ [6].

The edge feature is defined as $d_{rel}^{(i,j)} = \frac{\sqrt{(u_i - u_j)^2 + (v_i - v_j)^2}}{a_i + a_j}$, the relative distance between crater pairs. An edge is established between two craters if their relative distance is less than eight. We use relative values for the area and the distance because the absolute crater area and inter-crater distance are affected by the camera pose and imaging geometry. After the GAT layers, the aggregated node representations are passed through a linear classifier to produce the logit scores for the final crater IDs.

B. Crater Identification via teTRA (CITRA)

CITRA is based on the TETRA SID mechanism [5]. Fig. 2 summarizes the resulting pipeline. TETRA utilizes a group of four stars as a pattern for identification. Their six pairwise center distances are sorted in ascending order and normalized by the largest to form a key $r = (d_1/d_6, \dots, d_5/d_6) \in [0, 1]^5$ (d_6 is the longest) that is invariant to the report order of the detector and to any similarity transform of the image. The key is binned and stored in a hash table, so identification is a lookup rather than a search over the catalog. A nadir-pointing camera viewing a locally flat patch maps the ground to the frame by a similarity whose rotation equals the yaw and whose translation is given by the sub-camera point. We build the database from the filtered 13,574-crater catalog (detailed in Sec. III-A) by taking the 40 nearest neighbors of each

anchor and forming every three-combination with it, yielding 38,079,050 unique patterns.

The first modification addresses off-nadir viewing. A tilted camera maps the ground plane to the image by projectivity rather than similarity, which does not preserve the edge length ratios. As a lander can estimate its attitude from an IMU and a star tracker [9], we can easily warp each frame with a single fixed matrix per off-nadir angle that is independent of altitude, trajectory and catalog. Rim ellipses are warped by the conic rule $C' = H^{-T}CH^{-1}$ rather than by transforming their centers, since the projection of a crater center is not the center of the projected ellipse.

The second modification exploits the fact that craters carry size unlike stars. We extend the 5-D key with three sorted crater size ratios, which are invariant under the same similarity transformation as the original 5-D key. The resulting 8-D key is binned at a higher tolerance on the area dimensions than the edge dimensions, since size estimates are far noisier than centroids. Therefore, the area dimensions act as a coarse retrieval filter rather than a fine measurement.

At runtime the four-combinations of the eight largest detections are tried in order of size. Retrieved candidates pass a recomputation of the ratio bounds, from which the sub-camera point and yaw are recovered. A candidate is accepted only after the database is reprojected and TETRA’s binomial mismatch-probability test is passed.

III. EVALUATION

A. Evaluation Dataset

We evaluate our methodologies on a descent sequence synthesized using Lunar/LROC QuickMap [10], which utilizes pre-registered multi-scale lunar surface data. The trajectory targets the lunar south pole region and consists of 100 frames of 1024×1024 px with altitude decreasing from 100 km to 30 km. This lower bound is set by the 50 m/px resolution of the underlying DEM, below which the synthesized image may include spatial artifacts. Three series are generated from this single trajectory with off-nadir angles of 0° , 15° , and 30° to test the robustness to the viewing geometry.

We use the Robbins crater catalog [11], which provides broad coverage of lunar craters, to construct the CITRA database, train CIGNN, and reproduce the craters in the QuickMap dataset. Craters from the Robbins catalog are filtered to latitudes $\leq -77^\circ$, diameters ≤ 30 km, rim arc coverage $\geq 90\%$ and ellipticity ≤ 1.5 , yielding 13,547 craters in total. The ground-truth observations consist of every catalog crater whose center projects inside the frame with a margin, each with its full rim ellipse parameters.

B. Noise Application

To measure identification performance independently of any particular detector, the crater detection stage is bypassed. Instead, cataloged rims are projected into the image plane to form the observation set, and the noise characteristics a detector would introduce are injected. Specifically, four noise types (drop, false, position in x , y , and size in a , b) are applied. These types of noise are widely used for

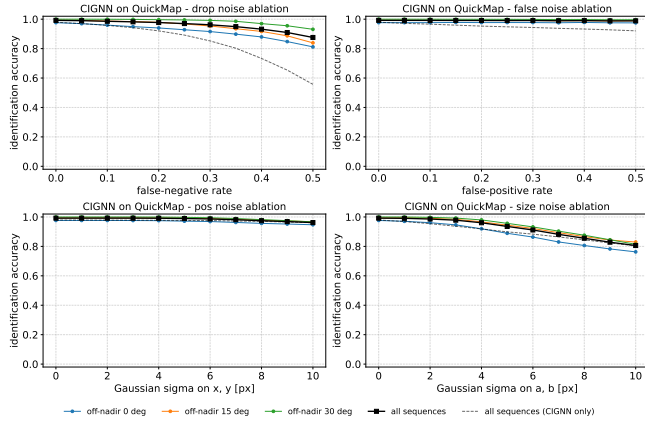


Fig. 3: CIGNN ablation on four types of noise on three sequences of the QuickMap dataset. CIGNN+PECAN accuracy is denoted by solid lines, while CIGNN-only accuracy is denoted by dotted lines.

ablation in CID research [7], [12] since they frequently occur in the crater detection step. In the ablation study, the drop and false-positive noise levels are varied from 0% to 50%, while the standard deviations of the position and size noise are varied from 0 to 10px in 10 steps. For the fixed-noise experiments used to compare CID performance, we adopt $p_{fn} = 0.316$, $p_{fp} = 0.43$, based on the performance of the Mask R-CNN-based CDA in [13], together with $\sigma = 0.5$ px, following the noise level used by Christian et al. [1] for CID validation.

C. Ablation Study

Fig. 3 presents the CIGNN (dotted lines) and CIGNN+PECAN (solid black lines) pipelines. CIGNN follows a robustness ordering close to that of SIGNN, being most robust to position noise, then false, size, and drop, against SIGNN’s position, false, drop, and magnitude. The last two are reversed mainly because size noise is larger in absolute terms and additionally perturbs the relative distance used for edge construction. Adding PECAN helps most under drop and false noise, since its inlier selection discards spurious craters and the PIE formulation recovers the remaining correspondences from a single correct identification. It helps least under size noise, where a perturbed rim yields an incorrect PIE pose and hence an inaccurate reprojection of the catalog, leaving size noise the hardest case for both stages.

Fig. 4 shows the corresponding ablation experiment for CITRA, whose robustness ordering is close to the reverse of CIGNN’s. CITRA is by far the most tolerant to size noise, losing only little accuracy across the full sweep. In the case of drop and false noise, it degrades roughly linearly. Position noise is the dominant failure case. Accuracy holds up to about 2 px and then collapses dramatically until about 0.1. The asymmetry follows from the key itself. The five edge dimensions are computed from the centroid distances and binned with a tight tolerance. Therefore, a centroid perturbation moves the pattern out of its bin and the lookup cannot return the correct IDs. On the other hand, the three

area dimensions are binned coarsely, so it can absorb the size perturbation.

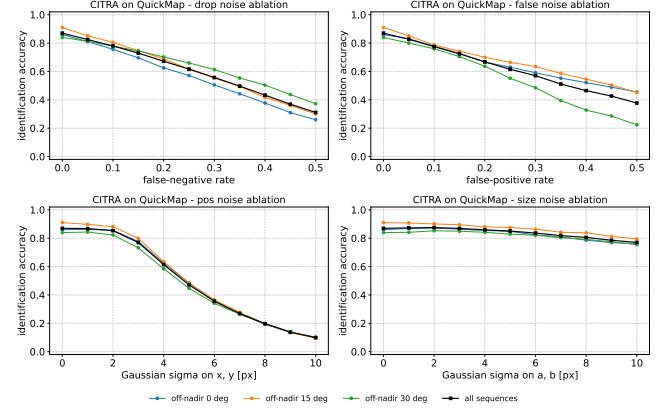


Fig. 4: CITRA ablation on four types of noise on three sequences of the QuickMap dataset.

D. Per-scenario Evaluation

Table I presents the identification accuracy and computation time of the baseline PECAN method and the proposed algorithms for each QuickMap scenario. The reported accuracy and computation time are the mean and standard deviation over 30 repeated runs of each scenario. PECAN requires a relatively long computation time because it sequentially pairs each image crater with catalog craters and searches through the entire catalog. However, in principle, once a reliable crater correspondence is successfully established, PECAN can recover the remaining correspondences and achieve a very high identification rate. CIGNN performs inference on an RTX 5090 and requires an average of approximately 2.1 ms per frame, making it the fastest among the four algorithms. When PECAN is combined with CIGNN, the identification rate improves by roughly 35% relative, while the computation time increases by roughly 29 ms. Thus, when comparing PECAN, CIGNN, and CIGNN+PECAN, PECAN provides the highest accuracy, CIGNN offers the best real-time performance, and CIGNN+PECAN can be regarded as a compromise between identification accuracy and computational efficiency.

Table I also reports CITRA over the three off-nadir series. Accuracy peaks at 15° (0.507 ± 0.030), while nadir and 30° are close to each other (0.436 and 0.418). Also, accuracy on solved frames varies by less than 0.05 across the series. Therefore, performance is determined by how often a frame is identified, not by how well. A frame is either identified nearly completely or not at all.

CITRA processes a frame in 52.0 ± 6.8 , 61.0 ± 6.3 and 89.5 ± 8.5 ms on average at 0°, 15° and 30° on a single CPU core, and even the slowest series averages under 100 ms per image. The reason for the bounded computation time is that patterns are formed only from the eight largest detections, so CITRA issues at most $\binom{8}{4} = 70$ lookups per frame. The mean detection count per image rises from 16.0 at nadir to 27.1 at 30°, while lookup rate rises only from 74% to 97% of the fixed ceiling.

TABLE I: Comparison of CID accuracy and computation time across three scenarios. Values are the mean $\pm$ standard deviation over 30 repeated runs of the 100-frame trajectory. CITRA and PECAN run on a single CPU core, while CIGNN inference runs on an RTX 5090. The best-performing results are shown in bold. *CITRA @ solved* restricts CITRA to the frames that return at least one correct label, and reports the accuracy and computation time over that subset.

Method	Off-nadir angle 0°		Off-nadir angle 15°		Off-nadir angle 30°	
	Accuracy	Computation time [ms]	Accuracy	Computation time [ms]	Accuracy	Computation time [ms]
PECAN	0.9821$\pm$0.0072	9800.1$\pm$622.5	0.9961$\pm$0.0030	9973.9$\pm$823.3	0.9995$\pm$0.0008	12804.2$\pm$566.7
CITRA	0.4362 $\pm$ 0.0293	52.0 $\pm$ 6.8	0.5073 $\pm$ 0.0299	61.0 $\pm$ 6.3	0.4178 $\pm$ 0.0377	89.5 $\pm$ 8.5
<i>CITRA @ solved</i>	0.9419 $\pm$ 0.0168	31.2 $\pm$ 7.3	0.9302 $\pm$ 0.0142	32.3 $\pm$ 8.0	0.8985 $\pm$ 0.0159	49.1 $\pm$ 11.5
CIGNN	0.6652 $\pm$ 0.0238	2.191$\pm$0.486	0.6971 $\pm$ 0.0224	2.076$\pm$0.030	0.7048 $\pm$ 0.0149	2.231$\pm$0.046
CIGNN+PECAN	0.9018 $\pm$ 0.0128	20.138 $\pm$ 7.308	0.9455 $\pm$ 0.0131	25.685 $\pm$ 2.078	0.9622 $\pm$ 0.0092	49.844 $\pm$ 2.680

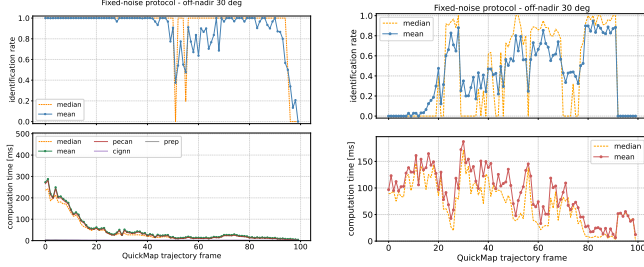


Fig. 5: Per-frame accuracy and computation time for CIGNN (left) and CITRA (right). Crater detection noises ($\sigma = 0.5$ px, $p_{fn} = 0.316$, $p_{fp} = 0.43$) are applied, and the evaluations are repeated 30 times. Mean and median values are calculated for each frame.

Fig. 5 resolves these averages along the trajectory for the 30° series. CIGNN loses accuracy around frames 50–60, where three or four visible craters are too few to form a graph with sufficient edges. On the other hand, CITRA loses accuracy over frames 0–20, where the frame is densest. The cost profiles are opposite as well with CITRA staying nearly flat while CIGNN+PECAN peaks in the early frames. The CIGNN trace itself is barely visible, confirming that PECAN dominates the combined cost and that supplying it with approximate IDs is what keeps that cost well below the figure reported in [7].

IV. CONCLUSION

In this work, we introduce the adaptation of SID algorithms to the CID problem. CITRA and CIGNN are based on two recent SID approaches: the deterministic TETRA algorithm and the learning-based SIGNN method, respectively. From the evaluation of these two methods on a simulated lunar lander dataset, we identify a key difference in their failure modes. CIGNN tends to fail when too few craters are visible to form a sufficiently informative graph, whereas CITRA degrades as the scene becomes denser since its candidate pattern pool is deliberately bounded, and is brittle to centroid noise because its hash bins fail discretely rather than gracefully. In terms of accuracy and computation time, PECAN achieves an identification accuracy of 0.99, outperforming CITRA (0.45) and CIGNN+PECAN (0.94). However, the computation times of CIGNN and CITRA are approximately 2.1 ms and 67.5 ms per frame, respectively, which are substantially shorter than PECAN’s 10.8 s. These results suggest that SID algorithms can be effectively adapted to CID with appropriate crater-specific modifications, as

demonstrated by CITRA and CIGNN, and that the proposed approaches may be better suited for real-time localization of lunar landers.

The current study is limited by the scale and diversity of the dataset and by the fact that the crater detection stage is bypassed. As future work, we plan to evaluate CITRA and CIGNN in a fully integrated pipeline with crater detection using more diverse datasets, such as CRESENT and SurRender-based datasets. Also, the Robbins catalog does not include crater elevation information. This can cause discrepancies between catalog craters projected onto the image and the actual crater rims [2], which should be calibrated in future work.

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